Laplacian Pyramid Dense Network for Hyperspectral Pansharpening

Wenqian Dong, *Member, IEEE*, Tongzhen Zhang, Jiahui Qu, *Member, IEEE*, Song Xiao, Jie Liang, *Senior Member, IEEE*, and Yunsong Li, *Member, IEEE*

***Abstract*— Hyperspectral (HS) pansharpening aims to create a pansharpened image that integrates the spatial details of the panchromatic (PAN) image and the spectral content of the HS image. In this article, we present a deep convolutional network within the mature Gaussian–Laplacian pyramid for pansharpening (LPPNet). The overall structure of LPPNet is a cascade of the Laplacian pyramid dense network with a similar structure at each pyramid level. Following the general idea of multiresolution analysis (MRA), the subband residuals of the desired HS images are extracted from the PAN image and injected into the upsampled HS image to reconstruct the high-resolution HS images level by level. Applying the mature Laplace pyramid decomposition technique to the convolution neural network (CNN) can simplify the pansharpening problem into several pyramid-level learning problems so that the pansharpening problem can be solved with a shallow CNN with fewer parameters. Specifically, the Laplacian pyramid technology is used to decompose the image into different levels that can differentiate large- and small-scale details, and each level is handled by a spatial subnetwork in a divide-and-conquer way to make the network more efficient. Experimental results show that the proposed LPPNet method performs favorably against some state-of-the-art pansharpening methods in terms of objective indexes and subjective visual appearance.**

***Index Terms*— Convolution neural network (CNN), hyperspectral (HS) pansharpening, Laplace pyramid decomposition technique, multiresolution analysis (MRA).**

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Wenqian Dong, Tongzhen Zhang, Jiahui Qu, and Yunsong Li are with the State Key Laboratory of Integrated Service Networks, Xidian University, Xi'an 710071, China (e-mail: wqdong@xidian.edu.cn; ztz_xd@163.com; jhqu@xidian.edu.cn; ysli@mail.xidian.edu.cn).

Song Xiao is with the Beijing Electronic Science and Technology Institute, Beijing 100070, China, and also with the State Key Laboratory of Integrated Service Networks, Xidian University, Xi'an 710071, China (e-mail: xiaosong@mail.xidian.edu.cn).

Jie Liang is with the School of Engineering Science, Simon Fraser University, Burnaby, BC V5A 1S6, Canada (e-mail: jiel@sfu.ca).

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I. Introduction

REMOTE sensing images are widely used in some remote sensing tasks. In addition, with the development of commercial software platforms, the demand for remote sensing images continues to grow. However, due to some physical limitations, such as onboard storage capacity, the images with high spatial, temporal, and spectral resolutions cannot be achieved by a single sensor. Fortunately, remote sensing image fusion provides an effective way to create the desired remote sensing images by fusing complementary information. According to the different purposes, remote sensing image fusion can be generally divided into the following several categories: spatiotemporal fusion, spatiospectral fusion, and spatiotemporal–spectral fusion [1]–[3]. Spatiotemporal fusion is designed to alleviate the problem of mutual restriction between the temporal resolution and the spatial resolution. Spatiotemporal–spectral fusion aims to realize the joint modeling of multitemporal, multispectral, and multiscale data. Spatiospectral fusion mainly solves the restriction problem between spatial resolution and spectral resolution. This article aims to study the spatiospectral fusion, where the panchromatic (PAN) image and the hyperspectral (HS) image are merged to obtain a fused image with high spatial and spectral performances. The PAN images imaged in a single electromagnetic spectrum have a high spatial resolution but low spectral diversity. The HS images characterized by dozens or even hundreds of spectral bands are complementary with PAN images in terms of spectral information and spatial information. HS pansharpening constitutes the sole possibility for integrating the spatial information of the high-resolution (HR) PAN image and the spectral content of the low-resolution (LR) HS image. The pansharpened product characterized by high spatial resolution and high spectral diversity improves the accuracy of its use in many applications.

In recent years, HS pansharpening has become a research hotspot in the field of remote sensing image processing. Many HS pansharpening algorithms have been developed and designed [4]–[6]. Vivone *et al.* [7] compared and analyzed the performance of some classical and emerging pansharpening methods in the literature. Initial efforts focus on two main families: 1) component substitution (CS)-based pansharpening algorithms that project HS image into a new space to separate the spectral and spatial components, substitute the separated spectral component with a PAN image, and generate the

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pansharpened image through the inverse spectral transformation and 2) multiresolution analysis (MRA)-based pansharpening algorithms that extract the missing spatial details of HS image from the PAN image and inject them into the HS image. The CS class contains many popular pansharpening algorithms, such as the Gram–Schmidt (GS) method [8], the principal component analysis (PCA) method [9], [10], and the intensity–hue–saturation (IHS) method [11]. Popular examples of the MRA class are smoothing filter-based intensity modulation (SFIM) [12] and the Laplacian pyramid [13]. In addition, some pansharpening methods that do not belong to these two families have been proposed in recent years. Among those, the sparse representation [14], [15], total variation penalization terms [16], and the parameter estimation-based Bayesian methods [17] also provide good pansharpening performance. By contrast, the CS and MRA algorithms are easier to implement without significantly reducing pansharpening performance, which is crucial for real-time processing [18], [19].

In recent years, deep learning has been successfully applied to many fields of image processing, especially in the field of image super-resolution, showing great potential in improving the quality of reconstructed images [20]. Dong *et al.* [20] proposed the first version of a deep convolution neural network (CNN)-based image super-resolution (SRCNN) method that shows better performance than some classic pansharpening methods. Inspired by the success of CNN in image super-resolution, Masi *et al.* [21] successfully applied SRCNN to pansharpening. Based on this, some residual learning-based pansharpening methods [22] were proposed, which provides the remarkable pansharpening performance achieved by the deeper CNN. Shen *et al.* [23] developed a spatial–spectral fusion framework by combining the deep learning and variational model. Deep learning has achieved remarkable success in pansharpening, particularly in the preservation of spatial details. For example, Scarpa *et al.* [24] proposed a target-adaptive CNN-based pansharpening (APNN) method, where, after the training, a target adaptive stage is implemented for recovering mismatches between training and testing datasets. They also proposed an improved version [25], which introduces a cross-scale learning strategy for CNN pansharpening models to better preserve the spatial details. Yang *et al.* [26] developed a deep network architecture for the pansharpening problem (PanNet), which trains the network parameters in the high-pass filtering domain to preserve the spatial structure. Wei *et al.* [22] have successfully developed a deep residual pansharpening CNN (DRPNN) that focuses on injecting the high-frequency detail into the multispectral image. Benzenati *et al.* [27] proposed a generalized Laplacian pyramid gain injection estimation based on CNN for pansharpening and achieved good spatial preservation performances by calculating the injection gains for each band.

Due to the different spectral responses and the complexity of the ground features, it is difficult for the classic pansharpening algorithms to formulate the relationship between the HR HS image, the PAN image, and the available LR-HS image. The high nonlinearity of CNN provides a new solution to this problem. The success of CNN-based methods proves the great potential of CNN in solving pansharpening problems. However, almost all CNN-based pansharpening methods are implemented by changing the network architectures of image super-resolution [28]. These methods essentially use the PAN image as an additional band of the HS image as the input to the network. The CNNs work like a black box, treating each band indistinguishable. Therefore, they cannot fully extract the spatial details of the PAN image, which plays an important role in pansharpening.

In this article, we propose a Laplacian pyramid network for HS pansharpening with the following advantages.

1) LPPNet has a clear physical interpretation of pansharpening, follows the general idea of MRA, and divides pansharpening into two continuous processes: detail extraction and reconstruction.
2) In the detailed extraction branch, we use the Laplacian pyramid to decompose the PAN image into multiple levels that can distinguish the details of different scales and build a simple detail extraction subnetwork for each level, which can help fully extract the depth features of different levels.
3) In the reconstruction branch, the subband residuals estimated at each level are injected into the respective level of HS image, while they are upsampled and input into the next subnetwork, which can help make full use of complementary details between different levels;
4) In order to learn more discriminative deep spatial features, the PAN images are processed by dense blocks, which encourages feature reuse, strengthens feature propagation, and makes the network more efficient.

The rest of this article is structured as follows. The MRA pansharpening methods are briefly reviewed in Section II. Section III presents the proposed pansharpening method. The experimental results are reported in Section IV. Section V draws the conclusion.

II. Related Works

In the MRA-based pansharpening methods, the contribution of the PAN image to the pansharpened product is realized by injecting the high-pass component of the PAN image into the HS image. A general formulation of MRA pansharpening methods is defined by [29], [30]

$$\widehat{\mathbf{H}}_k = \widetilde{\mathbf{H}}_k + \mathbf{G}_k \otimes \mathbf{D}_k \tag{1}$$

where subscript $k = 1, 2, \ldots, c$, c is the number of bands in the HS image, $\widetilde{\mathbf{H}}$ denotes the HS image interpolated to the same size as the PAN image, $\widehat{\mathbf{H}}$ denotes the fused HS image with HR, $\mathbf{G}$ represents the injection gain, symbol $\otimes$ represents the element-by-element multiplication operation, and $\mathbf{D}$ is the extracted detail image, which is generally achieved by exploiting the difference between the PAN image and its LR version

$$\mathbf{D} = \mathbf{P} - \mathbf{P}_L \tag{2}$$

in which the LR version of the PAN image $\mathbf{P}_L$ can usually be obtained by an iterative decomposition. However, the estimation of $\mathbf{P}_L$ heavily depends on the selected filtering scheme. On the one hand, the design of the spatial filter makes the

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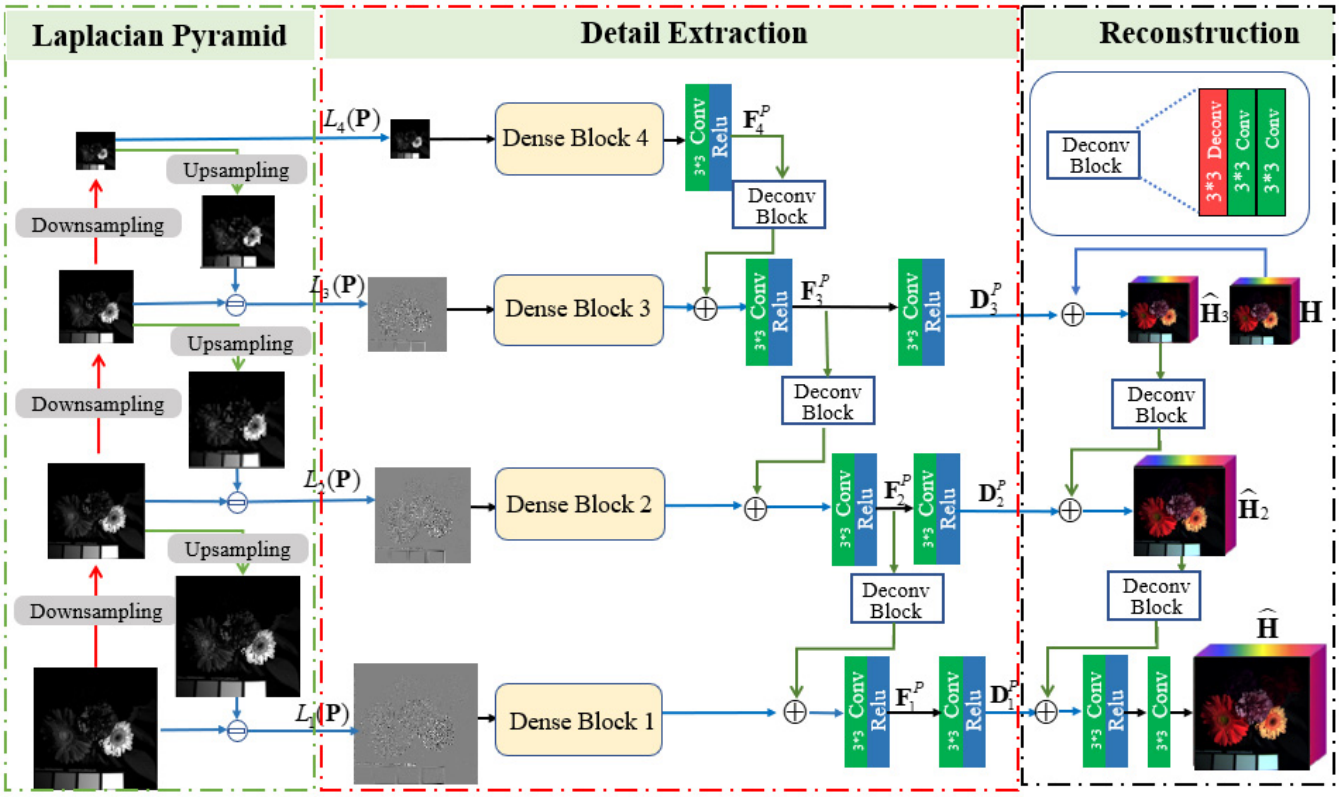


Fig. 1. Network architecture of LPPNet. The Laplacian pyramid transform is inside the green dotted area, the details extraction branch is inside the red dotted area, and the reconstruction branch is in the black dotted area. Black arrows indicate the convolution operation. Blue arrows indicate the element-by-element matrix addition and subtraction operators. Red arrows denote downsampling. Green arrows denote upsampling.

implementation of the MRA-based pansharpening methods more complicated. On the other hand, the mixing of the low-pass component of the PAN image will blur the edge and texture of the pansharpened image.

In the detailed injection step, defining an effective injection gain remains a challenging problem. In general, there are two widely used methods to define the injection gain. The first one is the global injection gain. The estimation of the injection gain is done globally, that is, per spectral band corresponds to a single injection coefficient for the whole image. Two common options for estimating the global gain are given as follows: 1) the injection gain $\mathbf{G}$ is a matrix with all elements equal to a constant, such as the HPF [11], ATWT [31], and MTF-GLP [32] methods or 2) the injection gain corresponding to each band is globally estimated. The global methods have the potential to greatly reduce computational complexity. However, the structural characteristics of HS images are usually complex. The global methods cannot deal with the redundant information between spectral bands and the difference of different land covers in the image. Thus, injection gains estimated by the global methods may cause spectral distortion.

The injection gains estimated by the context-adaptive methods are spatially variant. In this case, each pixel per band has a single injection coefficient, such as SFIM [12], HPM [33], MTF-GLP-HPM [34], and AWLP [35] fusion methods. The injection gain defined by the context-adaptive method helps maintain the local intensity contrast of the original image in the pansharpened image. However, they usually achieve better pansharpening performance at the expense of a high computational burden.

In summary, the MRA-based pansharpening methods decompose the PAN image into different levels that can distinguish large- and small-scale details through an iterative decomposition scheme and fully extract spatial information on different levels. Therefore, the pansharpening methods based on MRA are superior to other fusion methods in spatial quality improvement. However, the complexity of spatial filter design and injection gain estimation makes it difficult to implement. Fortunately, the high nonlinearity of the CNNs offers new solutions to formulate the relationship between the LR HS image, the PAN image, and the HR HS image. Following the general idea of MRA, this article proposes a new Laplacian pyramid network for pansharpening by combining CNN with the Laplacian pyramid technique.

III. Proposed Method

In many applications, the ideal resolution of an HS image is four times that of its original resolution. However, reconstruction artifacts often exist in the pansharpened image when the HS image is directly upsampled by a factor of four. To solve this problem, LPPNet uses the Laplacian pyramid network to extract details level by level and reconstructs HS image from coarse resolution to fine resolution. As shown in Fig. 1, the overall structure of the proposed LPPNet is a pyramid network that can be further divided into two main parts: 1) detail extraction and 2) reconstruction. Following the general idea of MRA, detail extraction and reconstruction are

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performed at each level. In detail extraction, the PAN image is first decomposed into a Laplacian pyramid. For each pyramid level, a special subnetwork is designed to extract the details from that level of the PAN image. In the reconstruction branch, the LR HS image is upsampled, and the extracted details are injected into the corresponding upsampled HS image level by level. The following contents describe these procedures in detail.

A. Laplacian Pyramid

LPPNet uses the Laplacian pyramid to decompose the PAN image into multiple levels and designs a simple subnetwork for each level. In each Laplacian pyramid decomposition, bilinear interpolation is used for both the downsampling and upsampling operations. The strategy of combining Laplacian pyramid decomposition and CNNs will allow the subnetwork to take full advantage of the sparsity of each level. The PAN image is first decomposed into its Laplace pyramid with four levels. The ith level of the Laplace pyramid $L_i(\mathbf{P})$ can be formulated as

$$L_i(\mathbf{P}) = G_i(\mathbf{P}) - \uparrow G_{i+1}(\mathbf{P}) \tag{3}$$

where the symbol $\uparrow$ represents the upsampling operation, $i = 1, 2, 3$, and $G_i(\mathbf{P})$ is the ith level of the Gaussian pyramid that can be achieved by downsampling $G_{i-1}(\mathbf{P})$, with $\mathbf{G}_1(\mathbf{P}) = \mathbf{P}$ and $\mathbf{G}_4(\mathbf{P}) = L_4(\mathbf{P})$.

B. Detail Extraction

The densely connected networks (DenseNets) [36] solve the problem of vanishing/exploding gradients by adding the output of any layer to the input of all subsequent layers. Inspired by the success of DenseNets in super-resolution [37], [38], the DenseNet-based multispectral pansharpening method has recently been proposed in [39]. The idea of DenseNets is particularly suitable for solving the generic pansharpening problem because it allows the shallow CNNs to extract more representative features in a feedforward way. However, the original DenseNet was designed to solve high-level vision tasks. The pansharpening methods based on DenseNets provide excellent pansharpening performance at the expense of introducing unnecessary calculations. In this article, we use DenseBlock to solve the pansharpening problem. By simplifying the number of convolution blocks in DenseBlock according to the requirements of different pyramid levels, the convergence rate of the model can be accelerated.

The whole detail extraction network is a cascade of DenseBlocks with a similar hierarchical structure. Specifically, the PAN image is decomposed into a multilevel Laplacian pyramid. A set of DenseBlocks with the same network structure and parameters is built to extract the features for each level.

1) DenseBlock: The main structure of the DenseBlock consists of several convolution layers for extracting features and a transition layer for controlling the number of channels. The dense connection pattern makes it great for encouraging feature reuse and alleviating the vanishing-gradient problem. Fig. 2 shows that the input of the lth convolution layer in the

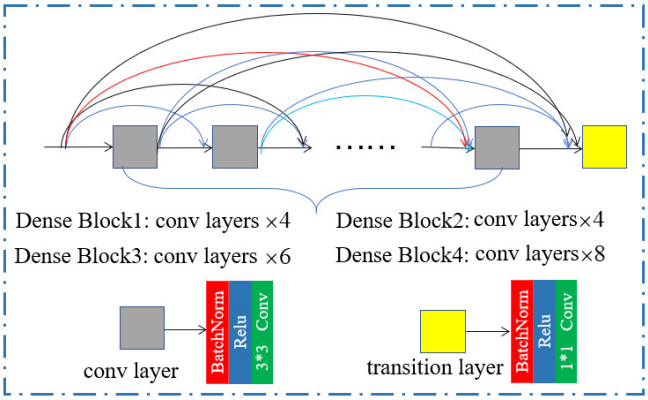


Fig. 2. Structure of the DenseBlock. The subnetwork designed for the four Laplace pyramids contains four, four, six, and eight convolution layers, respectively.

DenseBlock at the ith level receives the feature maps from all previous layers. Calculation in the lth convolution layer of the ith level is

$$\mathbf{P}_i^l = f_T\left(\left[L_i(\mathbf{P}), \mathbf{P}_i^1, \ldots, \mathbf{P}_i^q, \ldots, \mathbf{P}_i^{l-1}\right]\right) \tag{4}$$

where $0 \le q \le l-1$, $\mathbf{P}_i^q$ denotes the feature maps produced by the qth convolution layer of the DenseBlock at the ith level, and $[L_i(\mathbf{P}), \mathbf{P}_i^1, \ldots, \mathbf{P}_i^q, \ldots, \mathbf{P}_i^{l-1}]$ denotes the concatenation operation of the feature maps from all previous layers. $f_T(\cdot)$ is a consecutive operation consisting of BN, ReLU, and a 3×3 convolution. In each convolution layer, 32 filters of size $s \times 3 \times 3$ are used for generating 32 feature maps. For the lth convolution layer, $s = 1 + 32 \times (l-1)$.

2) Details Extraction: As shown in Fig. 1, the overall structure of details extraction branch is a cascade of the Laplacian pyramid dense network with a similar structure at each pyramid level. For each pyramid level, the feature maps of different levels are first extracted from the PAN image by a special DenseBlock. A transition layer in DenseBlock is connected to the last convolution layer and converts the feature maps into a c-channel feature maps, which will be fused with the feature maps generated at the previous level. The feature maps produced by the transition layer in the ith level of the Laplacian pyramid can be expressed as

$$\mathbf{P}_i^{d_i+1} = f_T\left(\left[L_i(\mathbf{P}), \mathbf{P}_i^1, \ldots, \mathbf{P}_i^{d_i}\right]\right) \tag{5}$$

where d_i is the number of convolution layers in the DenseBlock of the ith level. Empirically, four, four, six, and eight convolution layers are, respectively, adopted in the DenseBlocks of four Laplacian pyramid levels. The depth features of each level of the detail extraction subnetwork can be given as

$$\mathbf{F}_i^P = \begin{cases} f_{\text{cnn}}\left(\mathbf{P}_i^{d_i+1}\right), & \text{if } i = 4 \\ f_{\text{cnn}}\left[\mathbf{P}_i^{d_i+1} + \text{upsampling}\left(\mathbf{F}_{i+1}^P\right)\right], & \text{if } i = 1, 2, 3 \end{cases} \tag{6}$$

where the subscript i denotes the level index, $\mathbf{F}_i^P$ is the depth feature maps of PAN image at the ith level, the upsampling operation is performed by one 3×3 deconvolution layer and two 3×3 convolution layers, and the f_{cnn} is a $32 \times 3 \times 3 \times 32$

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convolutional layer followed by ReLU for nonlinearity. The extracted details at the ith level $\mathbf{D}_i^P$ can be formulated as

$$\mathbf{D}_i^P = f_{2\text{cnn}}\left(\mathbf{F}_i^P\right) \tag{7}$$

where $f_{2\text{cnn}}$ is the second convolution layer of $32 \times 3 \times 3 \times c$ at the ith level and c is the number of channel.

For the first three levels, the DenseBlock is followed by two convolutional layers. The output of the first convolutional layer is connected to the second convolutional layer for estimating the details of the current level i and is also upsampled and fused with the features of the level $i-1$ to extract the finer features. The second convolutional layer is used to reconstruct the output. ReLU is added between the first and second convolution layers for nonlinearity.

C. Image Reconstruction

At the third level of the reconstruction branch, the input image is the available LR HS image. The HS image is upsampled and fused with the predicted details from the details extraction branch level by level to reconstruct the HS image from coarse resolution to fine resolution. The product of level i is fed into the level $i-1$. Each reconstruction level can be given as

$$\begin{aligned}
\widehat{\mathbf{H}}_3 &= \mathbf{H} + \mathbf{D}_3^P \\
\widehat{\mathbf{H}}_2 &= \text{upsampling}\left(\widehat{\mathbf{H}}_3\right) + \mathbf{D}_2^P \\
\widehat{\mathbf{H}} &= f'_{\text{cnn}}\left\{ f_{\text{cnn}}\left[\text{upsampling}\left(\widehat{\mathbf{H}}_2\right) + \mathbf{D}_1^P\right]\right\}
\end{aligned} \tag{8}$$

where $\widehat{\mathbf{H}}$ is the pansharpened HS image, $\mathbf{H}$ is the available LR HS image, $\widehat{\mathbf{H}}_3$ and $\widehat{\mathbf{H}}_2$ represent HS images with details injected, which are extracted at levels 3 and 2, the upsampling operation is also performed by one 3×3 deconvolution layer and two 3×3 convolution layers, and f'_{cnn} and f_{cnn} are two adjacent convolution layers between which ReLU is added.

D. Loss Function

Let $\{\mathbf{H}^j, \mathbf{P}^j, \mathbf{H}_G^j\}_{j=1}^R$ represent the training dataset, where R denotes the number of training data, and $\mathbf{H}^j$, $\mathbf{P}^j$, and $\mathbf{H}_G^j$ represent the input LR HS image, the input PAN image, and the ground-truth HS image of the jth training sample, respectively. The proposed LPPNet predicts details at different levels. The ground truth of different levels can be obtained by downsampling the reference HS image; thus, the loss of different levels can be calculated. Since the square penalty does not work well at the edges of the image, mean squared error (mse) may produce oversmoothed results. Instead of minimizing the mse between the generated HR HS image $\widehat{\mathbf{H}}$ and the ground-truth HS image $\mathbf{H}_G$, we use ℓ_1 and correlation coefficient (CC) [40] to handle outliers. Based on the fact that the lower pyramid level contains the finer details, we use ℓ_1 and CC to train the first pyramid level to better preserve the details and use ℓ_1 as a loss function for all levels to achieve detail extraction and spectral fidelity. The overall loss function is given as

$$\mathcal{L} = \frac{1}{R}\sum_{j=1}^{R}\left\{\sum_{i=1}^{3}\mathcal{L}_{\ell_1}\left(\widehat{\mathbf{H}}_i^j, \mathbf{H}_{Gi}^j\right) + \beta \cdot \mathcal{L}_{(1-\text{CC})}\left(\widehat{\mathbf{H}}_1^j, \mathbf{H}_{G1}^j\right)\right\} \tag{9}$$

where R denotes the number of training sample, $\widehat{\mathbf{H}}_1 = \widehat{\mathbf{H}}$, $\mathbf{H}_{Gi}$ represents the image obtained by 2^{i-1} times downsampling the ground-truth HS image $\mathbf{H}_G$, with $\mathbf{H}_{G1} = \mathbf{H}_G$, and β denotes the coefficient used to balance the spatial and spectral fidelity. $\mathcal{L}_{\ell_1}$ represents ℓ_1 loss, and $\mathcal{L}_{(1-\text{CC})}$ denotes the $(1-\text{CC})$ loss which can be represented as (10), shown at the bottom of the page, where M is the number of pixels in one band of the pansharpened HS image, $\widehat{\mathbf{H}}_{k,m}^j$ and $\mathbf{H}_{Gk,m}^j$ denote the mth pixel on the kth of the jth sample, and $\mu_{\widehat{\mathbf{H}}}$ and $\mu_{\mathbf{H}}$ are the means of $\widehat{\mathbf{H}}_{k,m}^j$ and $\mathbf{H}_{Gk,m}^j$, respectively.

IV. Experimental Results and Analysis

We have performed some experiments to make sure that the proposed LPPNet works well. Experiments were conducted on four datasets consisting of two sets of Pavia Center [41]: Houston and Los Angeles. We compare the proposed LPPNet with some state-of-the-art methods in terms of objective indexes and visual appearance.

A. Databases and Experimental Setup

In this section, we give an introduction of three public datasets, including two synthetic datasets and one real dataset. The Pavia Center dataset that is a synthetic dataset is acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the city of Pavia, Italy, in the wavelength range of 430–860 nm. It contains 115 spectral bands, and 13 of them are water absorption and noise bands. The remaining 102 bands are used for experiments. The Pavia Center dataset is of size 1096×1096. However, some of these pixels do not contain any information. We intercept an image with a size of $960 \times 640 \times 102$ for experimentation. We divide the whole image into 12 patches of size $160 \times 320 \times 102$ without overlapping and randomly select nine patches for training and the remaining three patches for testing. To overcome the problem of the limited training samples, each patch is then divided into 21 small patches of size $160 \times 160 \times 102$ with stride 8. The sizes of the synthetic LR-HS and PAN images obtained through Wald's protocol are, respectively, $40 \times 40 \times 102$ and 160×160.

$$\mathcal{L}_{(1-\text{CC})} = \frac{1}{R}\sum_{j=1}^{R}\left\{1 - \frac{1}{c}\sum_{k=1}^{c}\left[\frac{\sum_{m=1}^{M}\left(\widehat{\mathbf{H}}_{k,m}^j - \mu_{\widehat{\mathbf{H}}}\right)\left(\mathbf{H}_{Gk,m}^j - \mu_{\mathbf{H}}\right)}{\sqrt{\sum_{m=1}^{M}\left(\widehat{\mathbf{H}}_{k,m}^j - \mu_{\widehat{\mathbf{H}}}\right)^2 \sum_{m=1}^{M}\left(\mathbf{H}_{Gk,m}^j - \mu_{\mathbf{H}}\right)^2}}\right]\right\} \tag{10}$$

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The Houston dataset was acquired by an ITRES-CASI 1500 HS sensor over the campus of the University of Houston and its neighboring urban areas. Each HS image comprises 144 bands covering the spectral range of 380–1050 nm, and each band contains 349×1905 pixels with a spatial resolution of 2.5 m. For the Houston dataset, a subimage with a size of $320 \times 1280 \times 144$ is used as the reference HS image. In the experiment, in order to get more training samples, the HS image and the PAN image are also partitioned into eight nonoverlapping patches, and each nonoverlapping patch is further split into 41 small patches. The sizes of the test HS and PAN images obtained by Wald's protocol are $40 \times 40 \times 102$ and 160×160, respectively.

The Los Angeles dataset was acquired over a port in the city of Los Angeles by the Hyperion sensor mounted on the Earth Observing One (EO-1) satellite. The EO-1 is equipped with a Hyperion sensor for capturing HS data and an Advanced Land Imager (ALI) sensor for capturing PAN data of the same scene. The HS image contains 242 spectral bands with a spatial resolution of 30 m; 97 spectral bands are uncalibrated and noisy bands. In the experiments, we used the remaining 145 pure spectral bands. The PAN image has a spatial resolution of 10 m. The original dimension of HS image is 150×150, and the original size of PAN image is 450×450. In the experiments, we intercepted small patches with the size of 54×54 and 162×162 in the upper left corner of the HS and PAN images, respectively, for testing, and then slide the remaining part into 144 overlapping blocks for training.

We implemented the proposed method with the PyTorch framework and trained them using the NVIDIA GTX 2080Ti GPU. In the training stage, the Adam optimizer [42] is employed. The parameters are updated with the first moment and the second moment in Adam. The first moment is the mean of the gradient, and the second moment is the variance of the gradient. The former moment is equivalent to momentum. The learning rate is set to 10^{-3}. The batch size is 16, and the optimization is terminated at the 300th epoch.

B. Compared Methods and Quality Assessment

In this section, we start with an introduction of the competing methods and then give the objective assessment methods used to verify the effectiveness of LPPNet.

1) Competing Methods: The performance of the proposed LPPNet is compared against ten widely accepted fusion algorithms: GS [8], GS adaptive (GSA) [43], guided filter PCA (GFPCA) [44], Bayesian sparse (BSF) [17], coupled nonnegative matrix factorization (CNMF) [45], SFIM [12], Hysure, deep attention network (DAN) [46], PanNet [26], and DRPNN [27]. GS and GSA are two classical CS-based methods. GFPCA is a representative of hybrid methods. CNMF, BSF, and Hysure are typical methods based on matrix factorization and the Bayesian model, respectively. SFIM is the most popular MRA-based pansharpening method. DAN, PanNet, and DRPNN are three recently proposed fusion frameworks based on a deep convolutional network. In the experiments, we use the optimal parameters or default parameters given by these methods.

2) Quality Assessment of Synthetic Dataset: In this article, the Pavia Center and Houston datasets are synthetic datasets. For the synthetic datasets, the HS images are available, and the PAN images are not available. Thus, the fusion performance of the synthetic dataset is evaluated at reduced resolution. Specifically, the available HS image is used as the reference image, and the simulated LR-HS and PAN images are generated according to Wald's protocol [47]. The simulated LR-HS image is created by applying a 9×9 Gaussian filter and then by downsampling every 4 pixels in the spatial dimension of the reference image. The simulated HR-PAN image is generated by averaging the bands of the visible range of the reference image. The simulated LR-HS and HR-PAN images are fused through different pansharpening algorithms to generate pansharpened HS images. The pansharpened HS image is compared with the reference HS image to verify the effectiveness of the pansharpening algorithms on the synthetic dataset. Four widely accepted indexes, including cross correlation [40], spectral angle mapper (SAM), root mse (RMSE), and erreur relative globale adimensionnelle de synthese (ERGAS) [48], are used to evaluate the objective quality of HS images reconstructed by different pansharpening methods. CC characterizes geometric distortion by calculating the similarity between the corresponding bands of the reference image and the fused image. The optimal value of CC is 0. The SAM index that reflects the spectral distortion is calculated by exploiting the spectral vector angle between the fused image and the reference image. The closer the value of SAM is to 0, the better the spectral fidelity of the pansharpened HS image. RMSE and ERGAS are all global indexes and measure the overall quality of the pansharpened image. The RMSE calculates the standard difference between the pansharpened image and the reference one. The smaller RMSE and ERGAS values mean that the pansharpened results are better.

3) Quality Assessment of Real Dataset: The Los Angeles dataset is a real dataset. The LR-HS image and the HR-PAN image are available. Different pansharpening methods are performed on the available HS and PAN images to create the pansharpened product. Since both the HS and PAN images are available, two assessment procedures, i.e., the reduced-resolution assessment and the full-resolution validation, are adopted to evaluate the objective quantitative quality of each method.

Reduced-Resolution Assessment: The available real HS image is used as the reference image, and the available HS and PAN images are degraded to produce the reduced-resolution HS and PAN images according to Wald's protocol [48]. The reduced-resolution HS and PAN images are fused by the competing pansharpening methods. CC, SAM, RMSE, and ERGAS are used to compare the pansharpened products with the reference HS image to achieve the quality assessment under the reduced resolution.

Full-Resolution Validation: Different from the reduced-resolution assessment, the full-resolution validation directly works on the available HS and PAN images. The objective fusion performance is quantified by the quality with no reference (QNR) image [49] index, which is calculated by two

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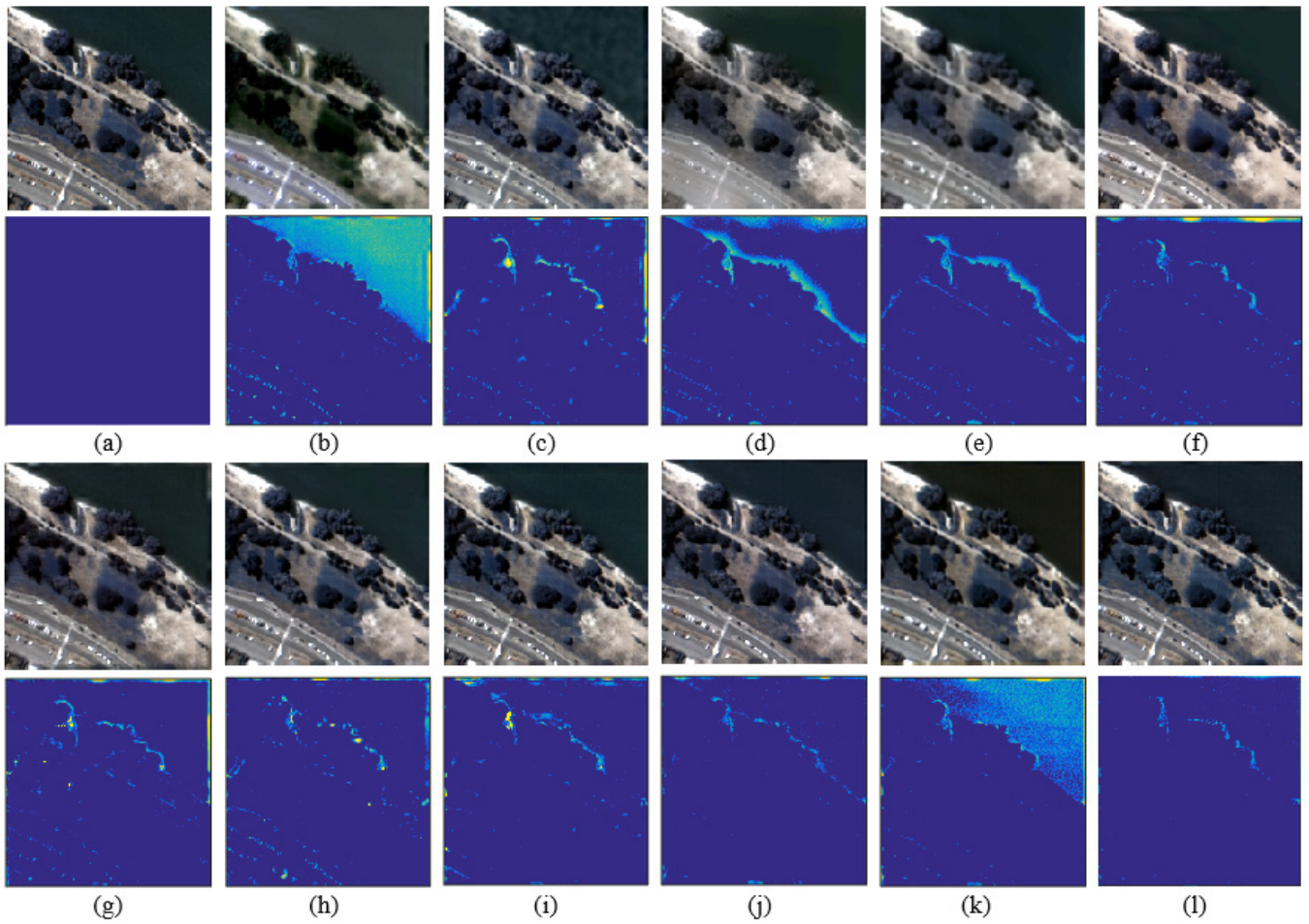


Fig. 3. First row: pansharpened images of the Pavia Center dataset. Second row: corresponding SAM maps (blue means small differences, and yellow means large differences) of the competing methods for the Pavia Center dataset. (a) Reference. (b) GS. (c) GSA. (d) GFPCA. (e) BSF. (f) CNMF. (g) SFIM. (h) Hysure. (i) PanNet. (j) DRPNN. (k) DAN. (l) LPPNet.

values: D_λ and D_s. The smaller the D_s, the better the spatial quality of the pansharpened image, and the closer the D_λ is to 0, the smaller the spectral distortion.

C. Experimental Results

Fig. 3 shows the visual results and the corresponding SAM maps of competing methods on the Pavia Center dataset. SAM map is obtained by calculating the SAM value between the pansharpened image and the reference image at each pixel, which can effectively reflect the spectral distortion of the pansharpened image obtained by different pansharpening methods. Fig. 4 shows the visual results and the corresponding error images of simulated experiments applied to another Pavia Center dataset. From a comparison of color bias between the pansharpened results and the reference image, it can be seen that most of the methods can get satisfactory results. The GS and DAN methods suffer from severe spectral distortion at the water areas, which is demonstrated by the SAM maps and error images. For the cars on the road, obvious blur artifacts also appear in Fig. 3(d) generated by the GFPCA method. In the GS, GSA, CNMF, and SFIM results, there were obvious color distortions in the water, road, and vegetation areas. In the results of the GS and BSF methods, it is difficult to identify the target boundaries, which indicates that they produce different levels of fuzzy artifacts scattered throughout the image, especially in the vegetation and roof areas. The Hysure and DAN methods work well for injecting the spatial details but produce noticeable color distortion, especially in the smooth area in Fig. 3 and the roof in Fig. 4. The PanNet, DRPNN, and proposed LPPNet model can well suppress the artifacts by progressive reconstruction and effectively preserves the spectral information of the original image. Compared with PanNet and DRPNN, the proposed LPPNet model causes the smallest SAM and pixel difference for most areas and achieves a better visual effect. The reason is twofold: first, our proposed LPPNet model follows the general idea of MRA and injects the details *in toto* the interpolated HS image, which enables the spectral consistency. Second, the proposed LPPNet model is trained by using a combined ℓ_1 and CC loss. This enables each subnetwork to handle small errors. Therefore, the proposed LPPNet performs better in preserving the spectral information and injecting the details than other competing methods.

The objective quality indexes of the Pavia Center dataset are listed in Table I, with the best value for each index in bold. The performance of GS and GFPCA is relatively inferior to those competing ones with respect to all objective evaluation indexes, revealing the reason that the whole image

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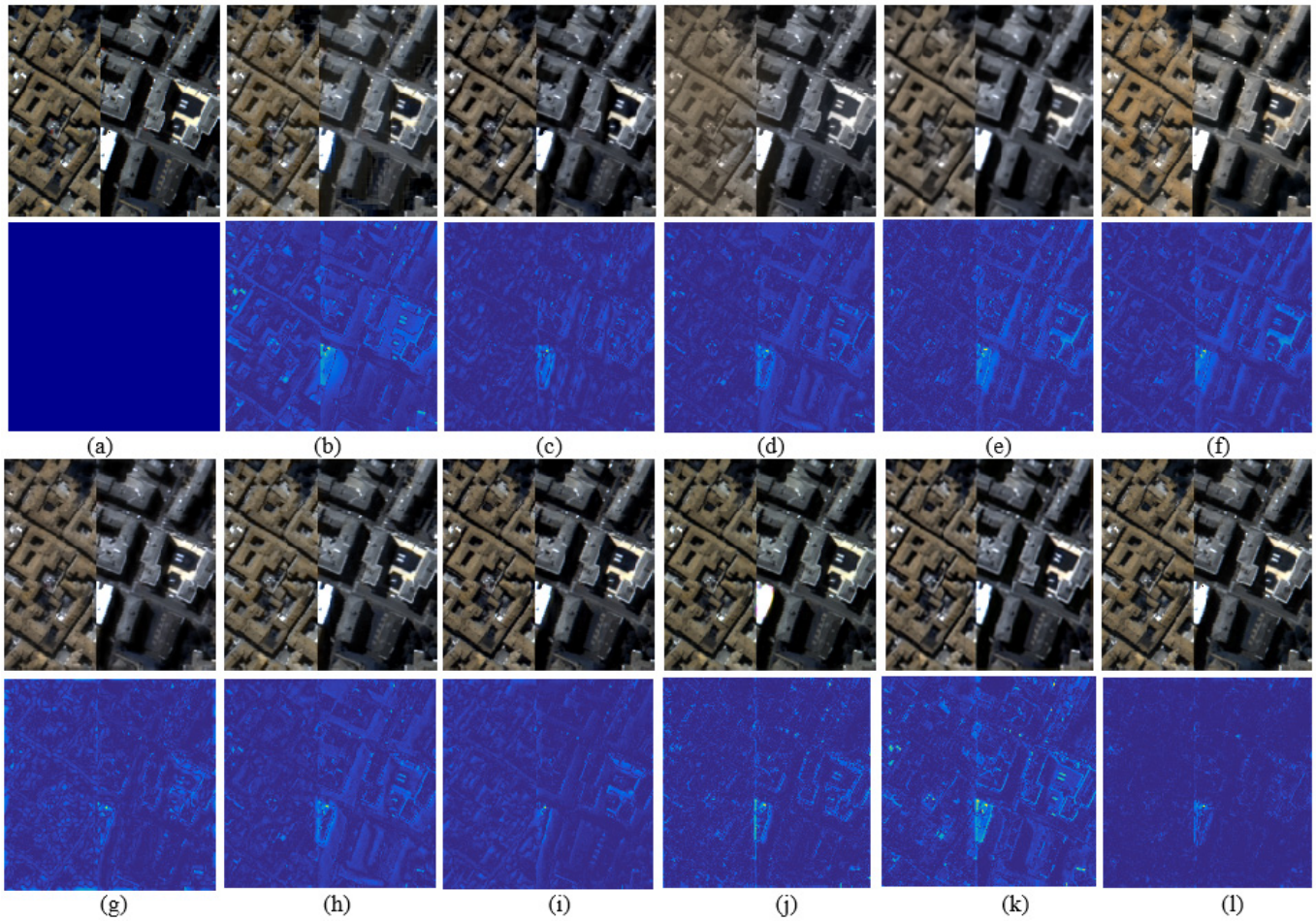


Fig. 4. First row: pansharpened images of another HS image from the Pavia Center dataset. Second row: corresponding error images (blue means small differences) of the competing methods. (a) Reference. (b) GS. (c) GSA. (d) GFPCA. (e) BSF. (f) CNMF. (g) SFIM. (h) Hysure. (i) PanNet. (j) DRPNN. (k) DAN. (l) LPPNet.

TABLE I
AVERAGE QUANTITATIVE ASSESSMENT FOR THE PAVIA CENTER DATASET

Method	CC	SAM	RMSE	ERGAS
GS	0.9265	9.5761	0.0398	7.6901
GSA	0.9357	7.9897	0.0314	5.8005
GFPCA	0.8951	8.7113	0.0441	8.2067
BSF	0.9358	8.3989	0.0316	6.3678
CNMF	0.9259	7.4225	0.0339	6.2678
SFIM	0.8988	7.9172	0.0390	7.4816
Hysure	0.9584	7.2519	0.0253	4.7958
PanNet	0.9704	6.4458	0.0200	4.0389
DRPNN	0.9537	6.1002	0.0263	5.0271
DAN	0.9684	8.9839	0.0205	4.4070
LPPNet	**0.9757**	**6.0166**	**0.0176**	**3.4047**

TABLE II
AVERAGE QUANTITATIVE ASSESSMENT FOR THE HOUSTON DATASET

Method	CC	SAM	RMSE	ERGAS
GS	0.8603	6.2270	0.0239	5.7810
GSA	0.8873	5.7160	0.0189	4.6935
GFPCA	0.8363	7.7363	0.0252	6.3024
BSF	0.8323	8.2025	0.0237	6.1518
CNMF	0.8926	5.7347	0.0191	4.7764
SFIM	0.8703	6.2834	0.0224	5.4571
Hysure	0.9149	4.7602	0.0167	4.0419
PanNet	0.9197	5.0829	0.0159	4.0399
DRPNN	0.9183	4.7284	0.0160	4.0398
DAN	0.9223	4.4520	0.0154	3.9695
LPPNet	**0.9357**	**4.2460**	**0.0147**	**3.9637**

is blurred and the color is obviously distorted on the roof. The disadvantage of BSF and DAN is obvious in terms of SAM. The PanNet and DRPNN methods can obtain very close results based on most objective quality indexes but slightly worse than the proposed methods. The proposed LPPNet method shows the best performance regarding CC, SAM, RMSE, and ERGAS, which is consistent with the visual assessment.

In order to compare the spectral distortion and the spatial distortion, we visualize and display the visual results of all the competing algorithms on the Houston dataset in Fig. 5. For clarity, we also show the error images that indicate the difference between the pansharpened images and the reference ones. As seen, GFPCA and BSF present an obvious lack of spectral information and spatial information. The HR-HS image provided by GFPCA has blur details on the surface of objects, which is caused by insufficient spatial detail injection. GS, SFIM, and PanNet suffer from different degrees of color distortion compared to the reference one, which can be caught easily in the error images. Compared with the reference image,

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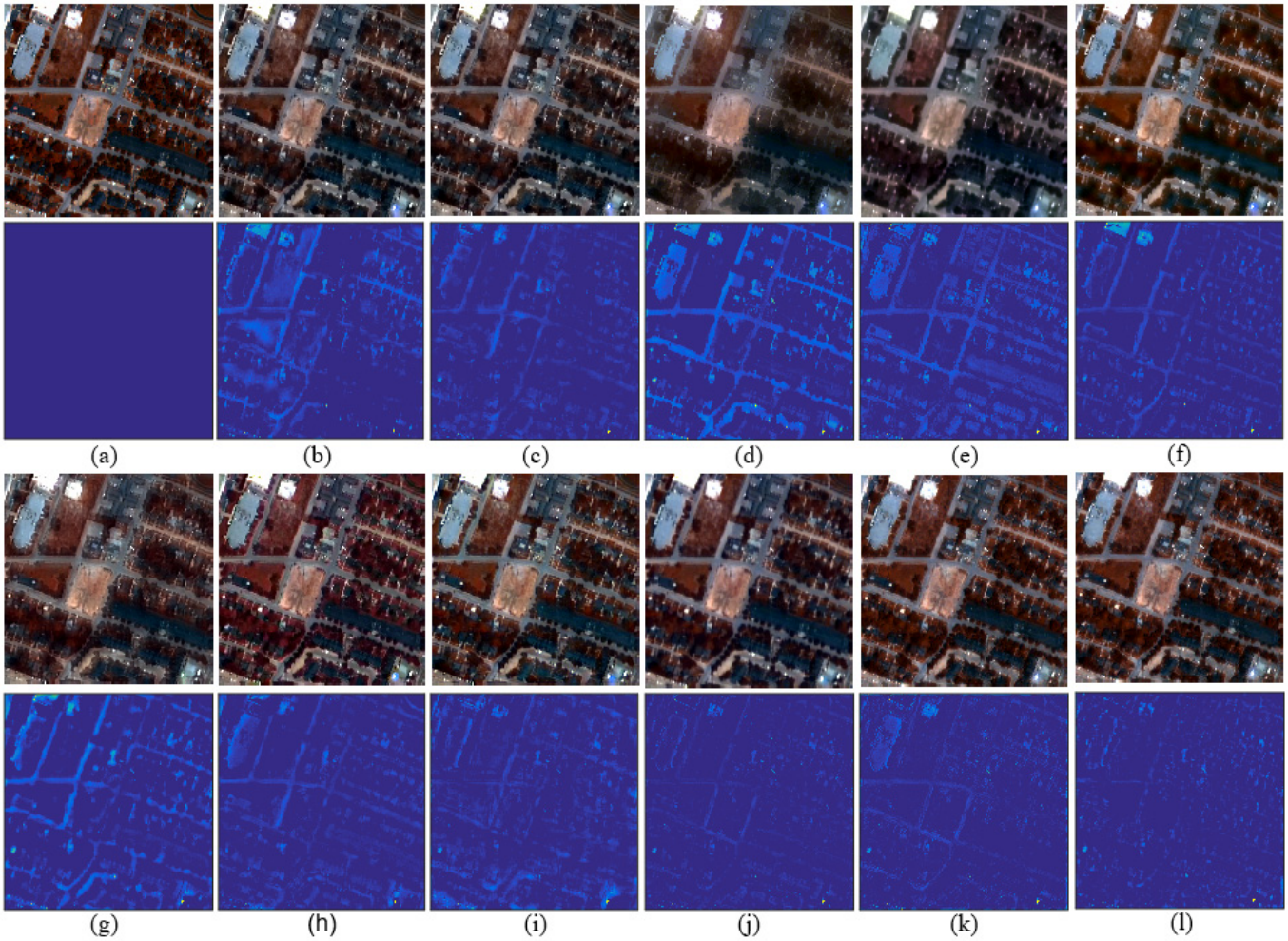


Fig. 5. First row: pansharpened images of the Houston dataset. Second row: corresponding error images (blue means small differences) of the competing methods for the Houston dataset. (a) Reference. (b) GS. (c) GSA. (d) GFPCA. (e) BSF. (f) CNMF. (g) SFIM. (h) Hysure. (i) PanNet. (j) DRPNN. (k) DAN. (l) LPPNet.

it can be seen that, although the HR-HS reconstructed by GSA and CNMF methods shows good spectral fidelity, they inject too much spatial information, which can be confirmed by the stripe artifacts existed on the edges. Deep learning has the powerful capability of feature learning and feature representation, and the CNN-based DRPNN, DAN, and proposed LPPNet methods can provide higher spectral fidelity and clearer detail than traditional methods. In particular, the method proposed in this article fully combines CNN and Laplace pyramid decomposition technology, which can fully extract the details of PAN images and, therefore, obtain HS images with high spectral fidelity and high spatial resolution.

Table II tabulates four widely used metrics for quantitative assessment in Fig. 5 to further evaluate the performance of competing methods, with the best values shown in bold. As can be observed, GFPCA and BSF show inferior performance than others in terms of all four quality evaluation indexes. The disadvantage of GS and SFIM is obvious in terms of CC, SAM, RMSE, and ERGAS, which gives the evidence for the former visual assessment. DAN and DRPNN achieve very competitive results, but the higher SAM values indicate that these two methods introduce more serious spectrum distortion than the proposed LPPNet. Our proposed LPPNet is superior to other competing methods, as it provides the biggest CC and the lowest SAM, RMSE, and ERGAS. Consistent with the subjective analysis, Table II further confirms the outstanding ability of LPPNet in balancing spectral preservation and spatial enhancement.

Fig. 6 shows the pansharpened results of the proposed LPPNet and competing methods on the Los Angeles dataset. The overall quality of most pansharpened images is significantly higher than that of the original LR-HS image, confirming the effectiveness of the pansharpening processing in improving the resolution of LR-HS images. However, some of these methods fail to restore the spatial details and spectral content properly. The pansharpened results provided by GS and Hysure suffer varying degrees of spectral distortion as revealed by the abnormal color of building metopes. The reconstructed HS image provided by the Hysure method also shows a slightly poor spectral fidelity. The degradation of spatial details can be observed in the products of GFPCA, SFIM, BSF, and PanNet, such as the edges of wall brick in Fig. 6. The buildings near the harbor are hard to recognize in the results of GFPCA, BSF, CNMF, SFIM, and PanNet methods, which indicates that these methods suffer from blur in weakly textured regions. By contrast, our proposed

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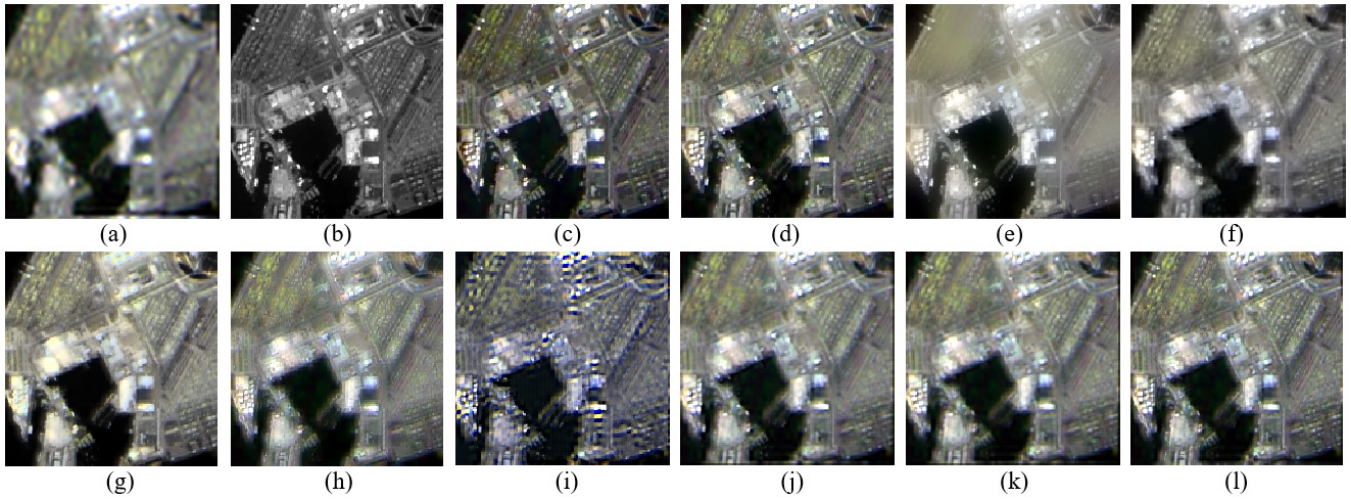


Fig. 6. Pansharpening results of the Los Angeles dataset with different methods. (a) Real HS image. (b) Real PAN image. (c) GS. (d) GSA. (e) GFPCA. (f) BSF. (g) CNMF. (h) SFIM. (i) Hysure. (j) PanNet. (k) DRPNN. (l) LPPNet.

TABLE III

AVERAGE QUANTITATIVE ASSESSMENT FOR THE LOS ANGELES DATASET

Method	GS	GSA	GFPCA	BSF	CNMF	SFIM	Hysure	PanNet	DRPNN	DAN	LPPNet
CC	0.8139	0.8358	0.8513	0.8687	0.8638	0.8384	0.8673	0.9385	0.9405	0.9439	**0.9446**
SAM	5.9436	5.2196	6.2644	5.5462	5.7755	5.2125	6.2953	**5.1275**	5.6724	6.8094	5.5198
RMSE	0.0734	0.0709	0.0681	0.0621	0.0632	0.0683	0.0747	0.0453	0.0517	0.0485	**0.0432**
ERGAS	8.3173	7.9913	7.7296	7.1589	7.1426	7.7468	8.4685	3.7248	4.9171	4.5328	**3.5809**
D_λ	0.0683	0.0750	0.0824	0.1182	0.0563	0.0584	0.0792	0.1014	0.0874	0.0640	**0.0561**
D_S	0.2145	0.1945	0.2263	0.1089	0.2035	0.1685	0.0736	0.0919	0.0899	0.1045	**0.0721**
QNR	0.7319	0.7451	0.7100	0.7858	0.7516	0.7829	0.8531	0.8160	0.8306	0.8382	**0.8758**

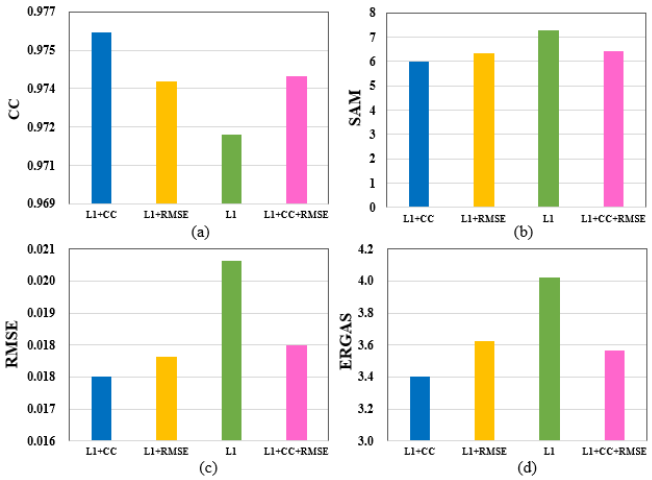


Fig. 7. Average quantitative results with different loss functions on the Pavia Center data set. (a) CC, (b) SAM, (c) RMSE, and (d) ERGAS.

LPPNet model outperforms the classical and state-of-the-art methods with comparable spatial restoration and negligible spectral distortions. This is because, after multiscale pyramid decomposition, each subnetwork only needs to deal with the corresponding scale, which helps to deal with real-world scenarios.

The quantitative results of different competing methods on the Los Angeles dataset are presented in Table III. It shows that the GS, GSA, and SFIM methods perform poor regarding CC. The Hysure, DAN, and GFPCA methods rank low in terms of SAM that accounts for the degree of spectral distortion. For the full-resolution validation, the BSF and PanNet methods obtain the poor performance regarding D_λ, and the disadvantage of the GS, GSA, GFPCA, and CNMF methods in improving spatial performance can be demonstrated by the value of D_s in Table III. By comparison, the experimental results shown in Table III demonstrate that the proposed LPPNet model achieves the best pansharpening performance among the competing methods with no obvious spectral distortion and artifacts, which validates the excellent fusion performance of our proposed LPPNet.

TABLE IV

AVERAGE QUANTITATIVE RESULTS WITH DIFFERENT WEIGHTING COEFFICIENTS ON THE PAVIA CENTER DATASET

Coefficient	CC	SAM	RMSE	ERGAS
0.01	0.9726	6.9223	0.0198	3.9435
0.1	0.9740	6.7555	0.0195	3.7057
1	**0.9757**	**6.0166**	**0.0176**	**3.4047**
10	0.9741	7.2773	0.0185	3.6211

D. Ablation Study

1) Loss Function: Because it does not excessively penalize large errors, the widely used l_1 loss has the ability to preserve detail and structure and, thus, becomes a more prominent measure in image enhancement. CC is based on local image features, such as contrast, brightness, and structures,

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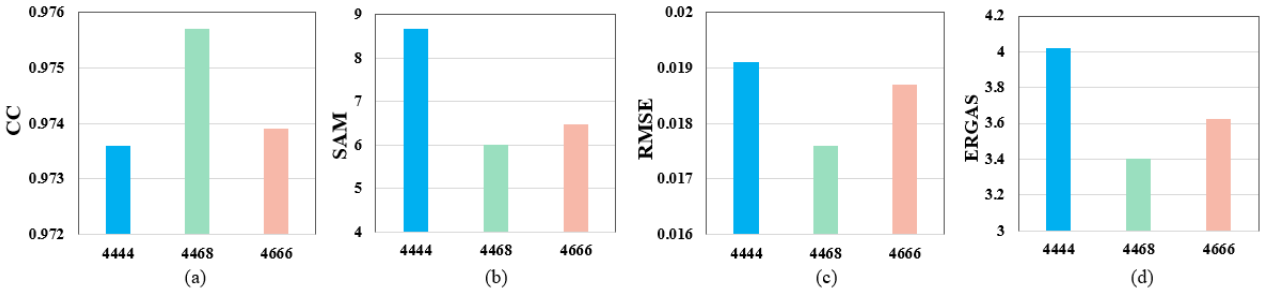


Fig. 8. Average quantitative results with different number of dense blocks on the Pavia Center data set. (a) CC, (b) SAM, (c) RMSE, and (d) ERGAS.

TABLE V
AVERAGE COMPUTING TIME OF DIFFERENT METHODS ON THE PAVIA CENTER DATASET

Method	GS	GSA	GFPCA	BSF	CNMF	SFIM	Hysure	PanNet	DRPNN	DAN	LPPNet
Time (s)	0.1566	0.1990	0.3911	25.3148	0.5299	0.1942	4.1848	0.0945	0.0199	0.0310	0.0541

to characterize the geometric distortion of pansharpened images. The human visual system is very sensitive to local image features. CC has the potential to create a visually pleasing result. Therefore, in this article, we use CC and l_1 as the loss function to guide the network training. Fig. 7 shows the comparative results of four objective evaluation indexes generated by using different loss functions. As can be seen in Fig. 7, our proposed loss shown in (9) generates comparable CC, SAM, RMSE, and ERGAS values. The proposed loss shown in (9) generates the best result followed by the combination loss of l_1, CC, and RMSE; the combination loss of l_1 and RMSE, finally, is l_1. Therefore, the loss shown in (9) is set as the default loss function in this article.

In (9), β is a weighting coefficient that balances the spatial and spectral fidelities. In order to study the influence of parameter β on the fusion performance, the proposed LPPNet is performed on the Pavia Center dataset with different β settings. Table IV shows the average quantitative assessment results with different β settings. As shown in Table IV, when the weighting coefficient β is set to 1, the proposed LPPNet achieves the best fusion performance. We have also tested the fusion performance of our proposed method on a large number of synthetic and real datasets and found that $\beta = 1$ also provides the best performance. Therefore, $\beta = 1$ is set to be the default parameter of the proposed LPPNet.

2) Conv Layer Number in DenseBlock: The performance of using different numbers of conv layers in DenseBlock of each subnetwork is compared in Fig. 8. Table VI reports the average quantitative results of different models on the Pavia Center dataset. As can be seen, we train three models with different conv layers in DenseBlock of each pyramid level: 4-4-4-4, 4-4-6-8, and 4-6-6-6. $n_a - n_b - n_c - n_d$ indicates that the DenseBlock in four pyramid levels, respectively, contains n_a, n_b, n_c, and n_d conv layers. As the number of conv layers increases from 4-4-4-4 to 4-4-6-8, a significant improvement is achieved by the proposed method. However, when the number of conv layers continues to increase from 4-4-6-8 to 4-6-6-6, the pansharpening performance degrades. This is because pansharpening focuses on learning low-level local features, and the nonlinear mapping of 4-4-6-8 conv layers in four pyramid levels can well solve the pansharpening problem.

TABLE VI
AVERAGE QUANTITATIVE RESULTS OF DIFFERENT MODELS ON THE PAVIA CENTER DATASET

Model	CC	SAM	RMSE	ERGAS	Time (s)
4-4-4-4	0.9736	8.6733	0.0191	3.8817	**0.0470**
4-6-6-6	0.9739	6.4687	0.0187	3.6428	0.0592
4-4-6-8	**0.9757**	**6.0166**	**0.0176**	**3.4047**	0.0541

Too deep network structure will lead to inefficient training. Therefore, 4-4-6-8 conv layers are set as the default setting.

3) Running Time: The average running time of different pansharpening methods on the Pavia center dataset is shown in Table V. All the classical pansharpening approaches are performed on CPU, and the deep-learning-based methods are performed on GPU. For 160 × 160-pixel image at PAN resolution, the average time cost of LPPNet is less than 0.1 s, which is acceptable compared with the classical pansharpening methods and deep-learning-based methods. The average time cost verifies the practicableness and efficiency of the proposed LPPNet. Table VI shows the time cost of different models on the Pavia Center dataset. As can be seen, the proposed 4-4-6-8 model achieves the best performance within acceptable time consumption.

V. CONCLUSION

In this article, a novel deep Laplacian pyramid network called LPPNet for HS pansharpening is proposed. The proposed LPPNet reconstructs the HR HS images in a coarse-to-fine manner. Specifically, the Laplacian pyramid is first adopted to decompose the PAN image into different levels that can differentiate large- and small-scale details. Then, a spatial dense network is designed for each level to extract the details. The Laplacian pyramids of PAN images are processed, while the HS image is upsampled. In addition, we design a special loss function for each level to train the subnetworks for better extracting details. The HR HS image is finally reconstructed at the bottom level of the pyramid. The proposed LPPNet is compared with ten traditional classical and recently proposed methods, and the results in widely accepted objective indexes and visual appearance demonstrate the superiority of the proposed LPPNet.

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Wenqian Dong (Member, IEEE) received the B.E. degree in communication engineering from Yantai University, Yantai, China, in 2014, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in June 2020.

She was a Visiting Scholar with Simon Fraser University, Burnaby, BC, Canada, from 2018 to 2019. She is a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. Her research interests include compressed sensing, machine learning, and hyperspectral remote sensing image processing.



Tongzhen Zhang received the B.E. degree in intelligence science and technology from Xidian University, Xi'an, China, in 2019, where he is pursuing the M.S. degree with the State Key Laboratory of Integrated Service Networks, Image Coding and Processing Center.

His research interests include hyperspectral image fusion and deep learning.



Jiahui Qu (Member, IEEE) received the B.S. degree in communication engineering from Yantai University, Yantai, China, in 2014, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in June 2020.

She was an Exchange Ph.D. Student with Mississippi State University, Starkville, MS, USA, from 2018 to 2019. She is a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. Her research interests include hyperspectral image detection, image fusion, neural networks, and deep learning.



Song Xiao received the M.S. degree in communication and information system and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2001 and 2004, respectively.

From 2006 to 2007, she was a Post-Doctoral Researcher with the Viterbi School of Engineering, University of Southern California, Los Angeles, CA, USA. She is a Professor and a Ph.D. Director of communication and information system with the State Key Laboratory of Integrated Service Networks, Xidian University. She has authored over 80 international journal articles and conference papers. Her research interests include image compression and coding, joint source-channel coding, multimedia transmission systems over wired/wireless networks, and compressed sensing.

Dr. Xiao is also a Council Member of the Shaanxi Society of Image and Graphics, the IEEE Multimedia Communication Technology Committee, and the IEEE Signal Processing Society. She is also the Secretary-General of the Image Application in Military and Civilian Integration (IAMCI) Professional Committee of the China Society of Image and Graphics.



Jie Liang (Senior Member, IEEE) received the B.E. and M.E. degrees from Xi'an Jiaotong University, Xi'an, China, in 1992 and 1995, respectively, the M.E. degree from the National University of Singapore (NUS), Singapore, in 1998, and the Ph.D. degree from Johns Hopkins University, Baltimore, MD, USA, in 2003.

From 1997 to 1999, he was with Hewlett-Packard Singapore, Singapore, and the Center for Wireless Communications, NUS. From 2003 to 2004, he was with the Video Codec Group, Microsoft Digital Media Division, Rockville, MD, USA. Since 2004, he has been with the School of Engineering Science, Simon Fraser University, Burnaby, BC, Canada, where he is a Professor and the Associate Director. In 2012, he visited the University of Erlangen-Nuremberg, Erlangen, Germany, as an Alexander von Humboldt Research Fellow. His research interests include image and video coding, multimedia communications, sparse signal processing, computer vision, and machine learning.

Dr. Liang received the 2014 IEEE TCSVT Best Associate Editor Award, the 2014 SFU Dean of Graduate Studies Award for Excellence in Leadership, and the 2015 Canada NSERC Discovery Accelerator Supplements (DAS) Award. He has served as an Associate Editor for the IEEE TRANSACTIONS ON IMAGE PROCESSING, the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT), the IEEE SIGNAL PROCESSING LETTERS, *Signal Processing: Image Communication*, and the *EURASIP Journal on Image and Video Processing*.



Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

He joined the School of Telecommunications Engineering, Xidian University in 1999, where he is a Professor. He is also the Director of the Image Coding and Processing Center, State Key Laboratory of Integrated Service Networks, Xidian University. His research interests include image and video processing, hyperspectral image processing, and high-performance computing.

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